# Distributed Systems Sample Test - CHAT GPT

## Instructions

Answer all questions clearly. Include diagrams when requested.

## Question 1: Consistency and CAP Theorem

(a) (2 points) Explain the difference between strong consistency and eventual consistency.

(b) (2 points) According to the CAP theorem, why is it impossible to guarantee all three CAP properties simultaneously?

(c) (1 point, True/False) A distributed system prioritizing availability and partition tolerance will always sacrifice strong consistency. *Circle True or False.*

## Question 2: Peer-to-Peer Networks

(a) (2 points) Describe two advantages and two disadvantages of peer-to-peer (P2P) architectures.

(b) (1 point, Fill in the blank) Distributed Hash Tables (DHT) are used in P2P networks primarily to achieve efficient _____.

## Question 3: Byzantine Fault Tolerance

(a) (2 points) Explain briefly what Byzantine faults are, and why they are challenging to handle.

(b) (2 points) How does the Byzantine Generals Problem illustrate the challenge of reaching consensus in distributed systems?

## Question 4: Election Algorithms

(a) (2 points) Describe the basic mechanism of the Bully Algorithm.

(b) (2 points) Provide one key advantage and one key disadvantage of using the Bully Algorithm compared to the Ring Election Algorithm.

## Question 5: HTTP and Cookies

(a) (1 point) Explain what a secure cookie flag does and who sets it.

(b) (1 point, True/False) Secure cookies ensure that cookies are only transmitted over HTTPS connections. *Circle True or False.*

## Question 6: Two-Phase Commit (2PC)

(a) (2 points) Outline the two main phases of the 2PC protocol.

(b) (2 points) Describe a scenario where 2PC might lead to a participant blocking indefinitely.

# Answer Key

## Question 1

(a) Strong consistency ensures all clients see the same data immediately after an update. Eventual consistency allows for temporary inconsistencies but ensures data convergence eventually.

(b) According to CAP theorem, in a network partition scenario, a system must choose between consistency and availability; thus, guaranteeing all three (consistency, availability, and partition tolerance) simultaneously is impossible.

(c) **True.**

## Question 2

(a) Advantages: Robustness, scalability; Disadvantages: Complexity in management, Security vulnerabilities.

(b) efficient **lookup and routing**

## Question 3

(a) Byzantine faults involve nodes failing unpredictably or maliciously, presenting incorrect or inconsistent data, challenging consensus protocols.

(b) The Byzantine Generals Problem illustrates the challenge by showing how faulty nodes can mislead others, making consensus difficult without sufficient redundancy and verification.

## Question 4

(a) The Bully Algorithm lets nodes initiate an election by sending messages to higher-ID nodes. If no response, the initiating node declares itself leader.

(b) Advantage: Simple and clear hierarchy. Disadvantage: High message overhead in large networks.

## Question 5

(a) Secure cookies ensure cookies are only sent over secure HTTPS connections. The server sets this flag.

(b) **True.**

## Question 6

(a) Two phases: (1) Voting (Prepare) phase, (2) Commit or Abort phase.

(b) If the coordinator crashes after sending a prepare message and before sending commit/abort, participants are left waiting indefinitely (blocked state).